Carmel Holy Word Secondary School

2024-2025 Term 2

Secondary 3 Computer Literacy

Practical Mock

Time: 45 Minutes

Name	
Class	
Class no.	
Date	
Marks	/ 50

1. Write your Name, Class and Class Number in the boxes provided.
2. Follow the instructions below to finish the examination tasks.
3. **<u>If you fail to upload your work, NO MARK will be given.</u>**

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1. **<u>ALL TASKS</u>** should be finished in **<u>VS Code</u>**.

2. Login to Teams and open a folder named "S3 Term 2 Practical Mock".

3. Download the files "task2_2425.py", "task3_2425.py" and "task4_2425.py" to your computer.

4. Submit your work to the suitable assignments in Teams.

Task 1 **(10 marks)**

Write a program to produce the output shown in Figure 1 for Task 1.

(a) **Create** a Python file in VS Code and **name** it “task1_2425.py”.

(b) **Ask** the user to input their name.

(c) **Ask** the user to input their age.

(d) **Calculate** the year they will turn 100 years old.

(e) **Print** the name and the calculated year (As **Figure 1** has shown).

(f) **Save the file as** “task1_2425.py” and **submit** it to Teams.

```
Enter your name: Eric
Enter your age: 20
Hello Eric! You will turn 100 years old in the year 2105.
```

Figure 1

Task 2 **(10 marks)**

Write a program to produce the output shown in Figure 2 for Task 2.

(a) **Download** “task2_2425.py” from Teams and **open** it in VS Code.

(b) **Create** an empty list to store the input number

(c) **Ask** user to input a number

(d) **Convert** the input string into an **integer.**

(e) **Complete** the if-elif-else condition.

- Check whether the input number is -1 or not
- If the input number is -1, stop the loop
- If the input number is greater than 0, add the number to the list
- If the input number is either -1 or greater than 0, print out the number and tell the user that the number is lower than 0 (As **Figure 2** has shown).

(f) **Print** the positive number saved in the list (As **Figure 2** has shown).

(g) **Save the file as** “task2_2425.py” and **submit** it to Teams

```
Enter a number (or -1 to stop): 12
Enter a number (or -1 to stop): 2
Enter a number (or -1 to stop): -9
-9 is lower than 0.
Enter a number (or -1 to stop): -1
[12, 2]
```

Figure 2

Task 3 **(15 marks)**

Write a program to produce the output shown in Figure 3 for Task 3.

(a) **Download** "task3_2425.py" from Teams and **open** it in VS Code.

(b) **Import** the library.

- Line 2: Speech Recognition library

(c) **Speech Recognition**: Set up the recognizer and the microphone

- Line 6: Set up the recognizer
- Line 7: Set up the microphone

(d) **Speech Recognition**: Detect user speech

- Line 17: Adjust the speech to cancel noises
- Line 18: Listen to user speech
- Line 21: Recognize user speech in English and output as text

(e) **Print** all the texts saved in the list

(f) **Save the file as** "task3_2425.py" and **submit** it to Teams.

```
Please say a sentence in English (say 'stop' to finish):
You said: hello nice to meet you
Please say a sentence in English (say 'stop' to finish):
You said: I hate you
Please say a sentence in English (say 'stop' to finish):
You said: stop
['hello nice to meet you', 'I hate you']
```

Figure 3

Task 4 **(15 marks)**

Write a program to display an image or a video.

(a) **Download** "task4_2425.py" from Teams and **open** it in VS Code.

(b) **Import** the OpenCV library

(c) **Loading an image**

- Line 13: Read the image file "face.jpg"
- Line 14: Show it as a window
- Line 15: Wait until any key is pressed to close the window

(d) **Loading a video**

- Line 19: Open the video file "video.mp4"
- Line 21: Read one frame from the video
- Line 22 - 23: Create an if condition to check if the video was playing or not; if not, stop video
- Line 24 – 25: Show the video

(e) **Save the file as** "task4_2425.py" and **submit** it to Teams.

End of Paper

Appendix: Modules and Functions API Reference Table

Module	pip install	Example of Import Statement
gTTS	`gTTS`	`from gtts import gTTS`
Playsound	`playsound==1.2.2`	`from playsound import playsound`
Google Cloud Translate	`google-cloud-translate==2.0.1`	`from google.cloud import translate_v2 as translate`
Speech Recognition	`SpeechRecognition`	`import speech_recognition as sr`
OpenCV	`opencv-contrib-python`	`import cv2`

Module	Description	Function Example
gTTS	Setup gTTS	`audio = gTTS(text="I Love You", lang="en")`
gTTS	Save TTS audio to a file	`audio.save("audio.mp3")`
Playsound	Play the audio file	`playsound("audio.mp3")`
Google Cloud Translate	Initialize the translation client	`client = translate.Client()`
Google Cloud Translate	Translate text between languages	`result = client.translate(text, source_language="en", target_language="zh")` `print(result["translatedText"])`
Speech Recognition	Create a Recognizer instance	`recognizer = sr.Recognizer()`
Speech Recognition	Initialize microphone input	`mic = sr.Microphone()`
Speech Recognition	Calibrate for ambient noise	`recognizer.adjust_for_ambient_noise(source)`
Speech Recognition	Listen and capture audio input	`audio = recognizer.listen(source)`
Speech Recognition	Convert audio to text using Google	`text = recognizer.recognize_google(audio, language="en")`
OpenCV	Read an image file	`image = cv2.imread("face.jpg")`
OpenCV	Open a video file or camera stream	`cap = cv2.VideoCapture("video.mp4")`
OpenCV	Read a frame from the video	`isTrue, frame = cap.read()`
OpenCV	Display the video frame in a window	`cv2.imshow('Video', frame)`
OpenCV	Wait for a key event (with delay)	`cv2.waitKey(1)`